DATA ITEM DESCRIPTION
Title: Model Federation Plan (MFP)

Number: DI-SESS-82560
AMSC Number: 10703
DTIC Applicable: No
Preparing Activity: 19
Applicable Forms: N/A

Approval Date: 20261001
Limitation: None
GIMFP Applicable: No
Project Number: SESS-2026-034

Use/relationship:

The Model Federation Plan (MFP) documents the contractor's technical approach for conducting, managing, and controlling the model federation effort on the program. It specifies how federated models will be integrated, governed, and maintained, and it identifies the goals, focus areas, methods, processes, tools, technologies, data, and resources that support successful federation. The primary objective is to enable federation of models within a program that provides value through integration, while also supporting cross-program federation where applicable.

In this document a model is any physical, mathematical, descriptive, logical, or otherwise abstracted representation of a system, object, phenomenon, or process. The term covers several model families that are typically used in Model-Based Systems Engineering, including, requirements models, architecture models, descriptive models, physics-based simulations, metamodels, and pattern libraries. The plan does not limit the federation to Systems Engineering models alone; analytic, developmental, lifecycle, descriptive models, and physics-based simulations are also included when they are part of the system-level representation and need to exchange data with other models.

Model federation is the disciplined practice of uniting models—each adhering to a common set of interfaces, governance rules, and reusable elements—into a governed ecosystem. The ecosystem provides defined interfaces, reusable elements, and governance mechanisms. Once united, a model federation can assist a program or system perform: Cross-Domain Impact Analysis, Technical Reviews, Verification and Validation, and Configuration Status Accounting.

A federated model is a model that conforms to the federation’s interface specifications, governance policies, usage patterns, and data exchange standards. Federated models may include requirements models, architecture models, simulation models, metamodel components, or other model types as determined by program needs. Not all models within a program are required to be federated; only those that benefit from cross-program integration or cross-domain data exchange need conform to federation standards.

This DID defines the format, content, and intended-use information for the federated model deliverables produced under the solicitation tasks. It ensures that every deliverable incorporates the required reusable profiles, libraries, or metamodel patterns and conforms to the interface and governance rules set out in the MFP. This DID should be used in conjunction with DI-SESS-82380, Model Based Systems Engineering (MBSE) Development Plan (MDP), and DI-SESS-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

82400, System Architecture Model (SAM). (Copies of this document are available online at https://quicksearch.dla.mil.)

Requirements:

1. **Reference Documents.** The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. **Format.** The delivered MFP shall be a document in the contractor's preferred format with the content specified in this DID. If requirements are tailored out of this DID, the resulting description shall contain a corresponding placeholder appended with "This content has been tailored out."

3. **Content**. The MFP shall include the following topics required for the successful development and implementation of model federation:

3.1. **Introduction.** This section shall provide a concise overview of the program's model federation and define its purpose, scope, management approach, and the assumptions/constraints that govern the model federation effort.

3.1.1. **Purpose and Objectives.** This subsection shall detail the intent behind implementing model federation within the program, outlining the benefits and anticipated outcomes and establishing the rationale for model federation. It shall also include the primary users of the model federation and their associated use cases.

3.1.2. **Federation Scope**. This subsection shall list each federated model's name, source, version, native modeling tool, data format and owner that will be included in the federation. In the case where constituent models are not under the direct control of the developer, this section shall also state the agreements, update frequencies, and trigger mechanisms by which those models are updated and synchronized.

3.1.3. **Governance.** This subsection shall describe the governance framework for the federation, including organizational roles and responsibilities, decision-making authority, data ownership, intellectual property boundary management, access control, and configuration control processes to ensure federation integrity and compliance.

3.1.4. **Schedule.** This subsection shall include a notional schedule for the model federation, including dates of major milestones and deliverables.

3.1.5. **Assumptions and Constraints.** This subsection shall identify the underlying assumptions and limiting constraints for tool availability, data formats, resource limits, intellectual property and proprietary data constraints, and regulatory requirements that affect goals, methods, processes, technology,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

data, and resources required for model federation.

3.2. **Approach.** This section shall detail the comprehensive approach for model federation, including the overall strategy. The following subsections address the technical, organizational, and operational dimensions necessary to integrate, manage, and sustain federated models throughout their lifecycle.

3.2.1. **Methodology.** This subsection shall provide a comprehensive overview of the methodology employed for model federation. It shall outline the structured workflow and processes for developing and integrating federated models, ingesting or linking legacy document-based technical baselines, quality assurance, cross-model traceability, data validation, data verification and stakeholder collaboration. This section should also address any workflows or processes put in place to mitigate or address security considerations. This subsection shall identify all modeling standards, style guides, glossary models, data exchange protocols, and other standardization artifacts that will be adhered to across the federation to ensure consistency and enable integration opportunities across organizations throughout the lifecycle.

3.2.2. **Architecture and Hierarchy**. This subsection shall define and describe the architecture of the model federation, which should reflect the architecture of the system or system of systems being represented. It shall include a visual representation and description of the federated model structure, illustrating organizational relationships between models.

3.2.3. **Environment.** This subsection shall define and describe the digital environment(s) in which federated models are developed, maintained, and accessed. This shall include identification of the software tools, computing platforms, data repositories, classification levels and supporting infrastructure used by each participating organization or system, as well as any shared or common infrastructure that enables federation across these environments. The subsection shall address ownership and responsibility for maintaining and sharing each modeling environment, recognizing that different modeling environments have distinct requirements for access, usability, and data transfer. Only environments supporting models identified in the Federation Scope that require cross-domain data exchange or cross-program integration need to be documented.

3.2.4. **Common Elements.** This subsection shall describe the types of common elements that will be used by multiple Federated Models. The subsection shall describe where common elements are stored, how they are accessed and shared, and who has version control over the elements. This shall include identification of any data dictionaries or pattern libraries utilized to standardize model elements definitions and reusable components.

3.2.5. **Scalability**. This subsection shall describe the approach for handling an

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

increasing number of federated models, large models, and the addition of new models. It shall address concerns related to large volumes of model data, anticipated model performance, support for multiple concurrent users, hardware or cloud infrastructure scaling triggers and other potential scaling challenges.

3.2.6. **Interoperability.** This subsection shall describe how the federated models exchange information. As the semantic foundation for this exchange, it shall define the overarching data ontology and metamodel structure, establishing the common vocabulary, data types, and logical relationships used across the federation. It shall identify each pair of interacting models, specify the data elements that are shared, and state whether the exchange is unidirectional or bidirectional. For each connection the description shall include the synchronization method, the explicit designation of the Authoritative Source of Truth (ASoT) for each shared data element to prevent data duplication, conflicts, and baseline divergence, the conflict resolution approach, and the standards or protocols used, prioritizing DoD-approved open data exchange standards. Additionally, it shall list the integration tools or middleware that will implement these standard interfaces and the format of the exchanged data.

3.2.7. **External Federation.** This subsection shall describe the approach for federating models between the program and external sources, including mission partners, supporting systems, and enterprise frameworks. For each external federation relationship, this subsection shall specify: the model source and maintaining organization; the interface specifications and data exchange protocols; governance mechanisms for access, validation, and version control; agreements, update frequencies, and trigger mechanisms by which those models are updated and synchronized; and security, classification, conflict resolution, and proprietary data handling requirements.

3.2.8. **Operational Application and Workflows.** This subsection shall detail how the model federation will be utilized by engineering and management teams to execute program activities. It shall describe the specific workflows and processes that leverage the federated ecosystem.

3.3. **Terms and Definitions**. This section shall define and describe an alphabetical listing of all terms, acronyms, and their definitions as used in the MFP.

END OF DI-SESS-82560

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.